

Compress-Then-Explain on Tabular Data

ecom-offers · TabReD | Part 2: Preliminary Results

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Agenda

1. Dataset & Setup

ecom-offers (TabReD), XGBoost baseline, AUC = 0.56

2. Research Questions

RQ1-RQ4 framing

3. RQ1 — Fidelity vs Size

CTE vs Random MAE across background sizes

4. RQ2 — Temporal Drift

Explanation stability across 5 time windows

5. RQ3 — Train vs Val

How explanations differ between splits

6. RQ4 — Break-even

When does the CTE build cost pay off?

7. Conclusions & Next Steps

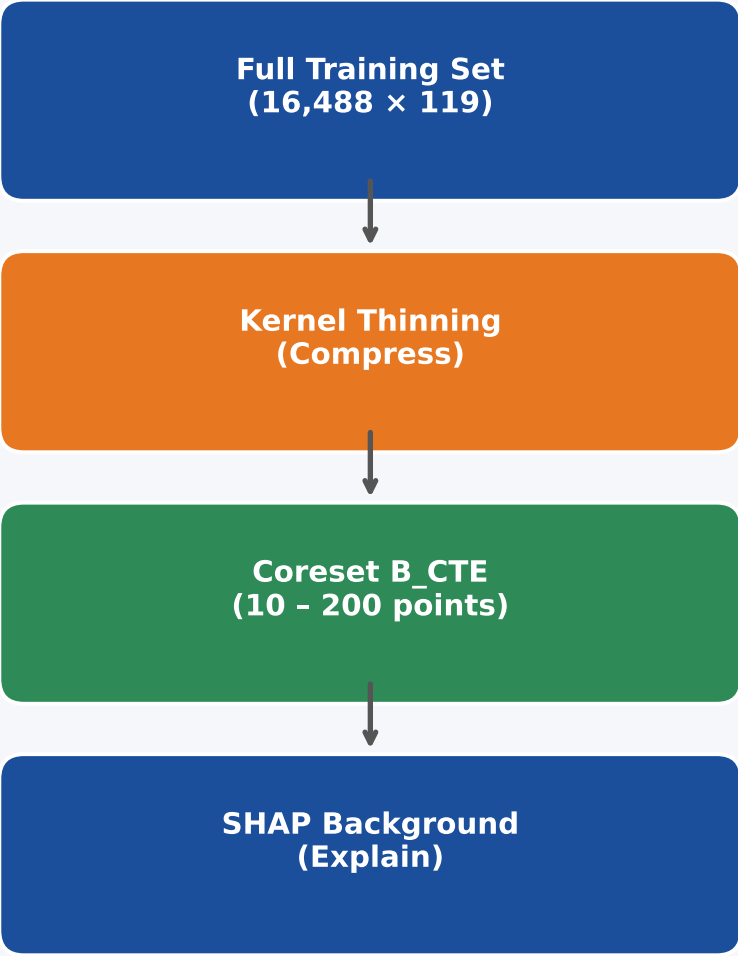
Plan for full dataset run

Dataset & Experimental Setup

Dataset Stats

CTE Pipeline

Property	Value
Dataset	ecom-offers (TabReD)
Task	Binary classification (offer redemption)
Train size	16,488 samples
Val size	3,598 samples
Features	119
Class balance (train / val)	23% / 37% positive
Model	XGBoost (default params)
AUC (val)	0.56
Background ref.	16,488 samples (full train)



Research Questions

RQ1

Fidelity-Efficiency Trade-off

Does CTE achieve lower SHAP approximation error than random sampling at the same background size? Is there a speedup?

RQ2

Temporal Stability of Explanations

Do feature importances shift across chronological validation windows? How well does a fixed CTE background track these shifts?

RQ3

Train vs Validation Discrepancy

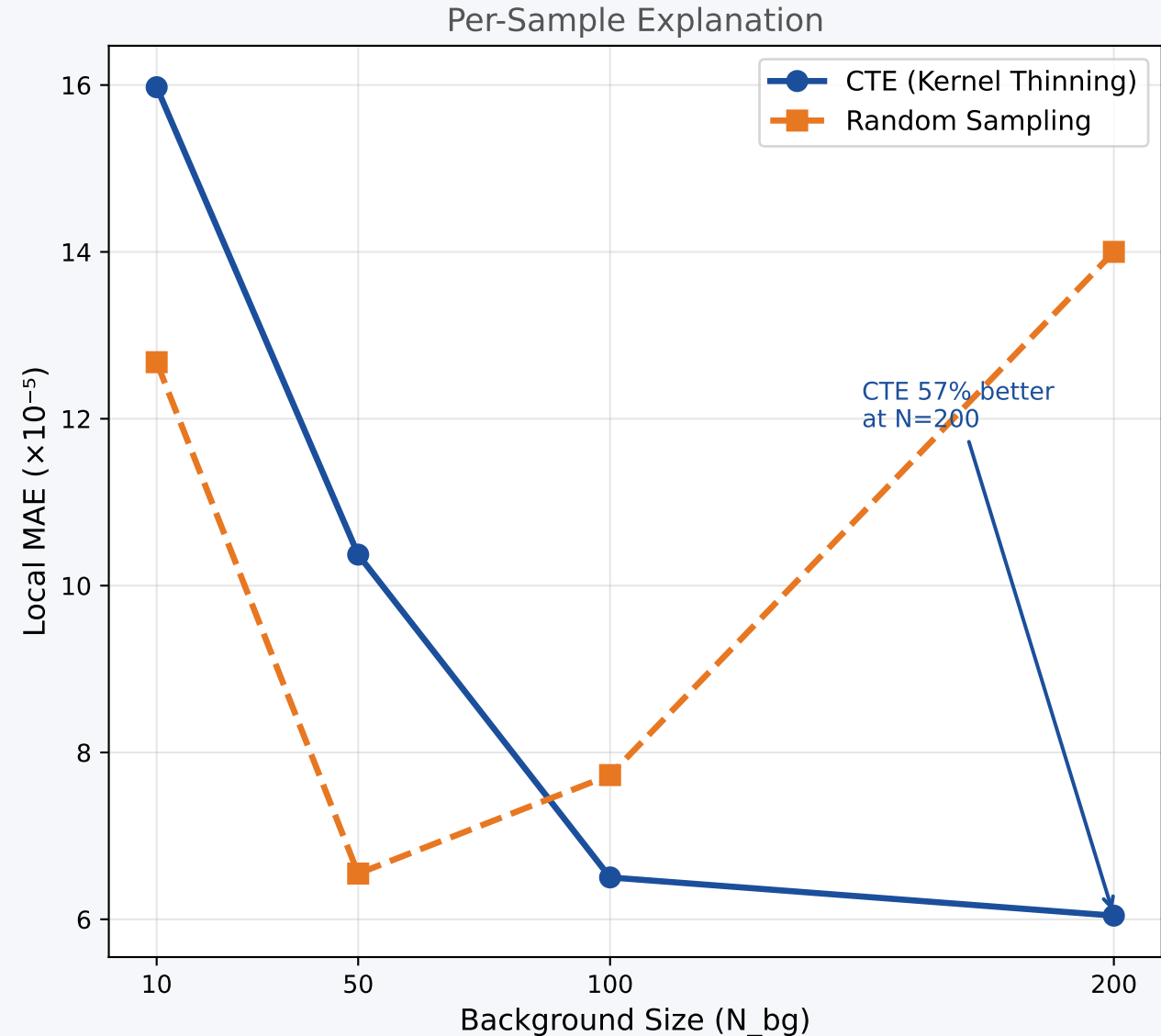
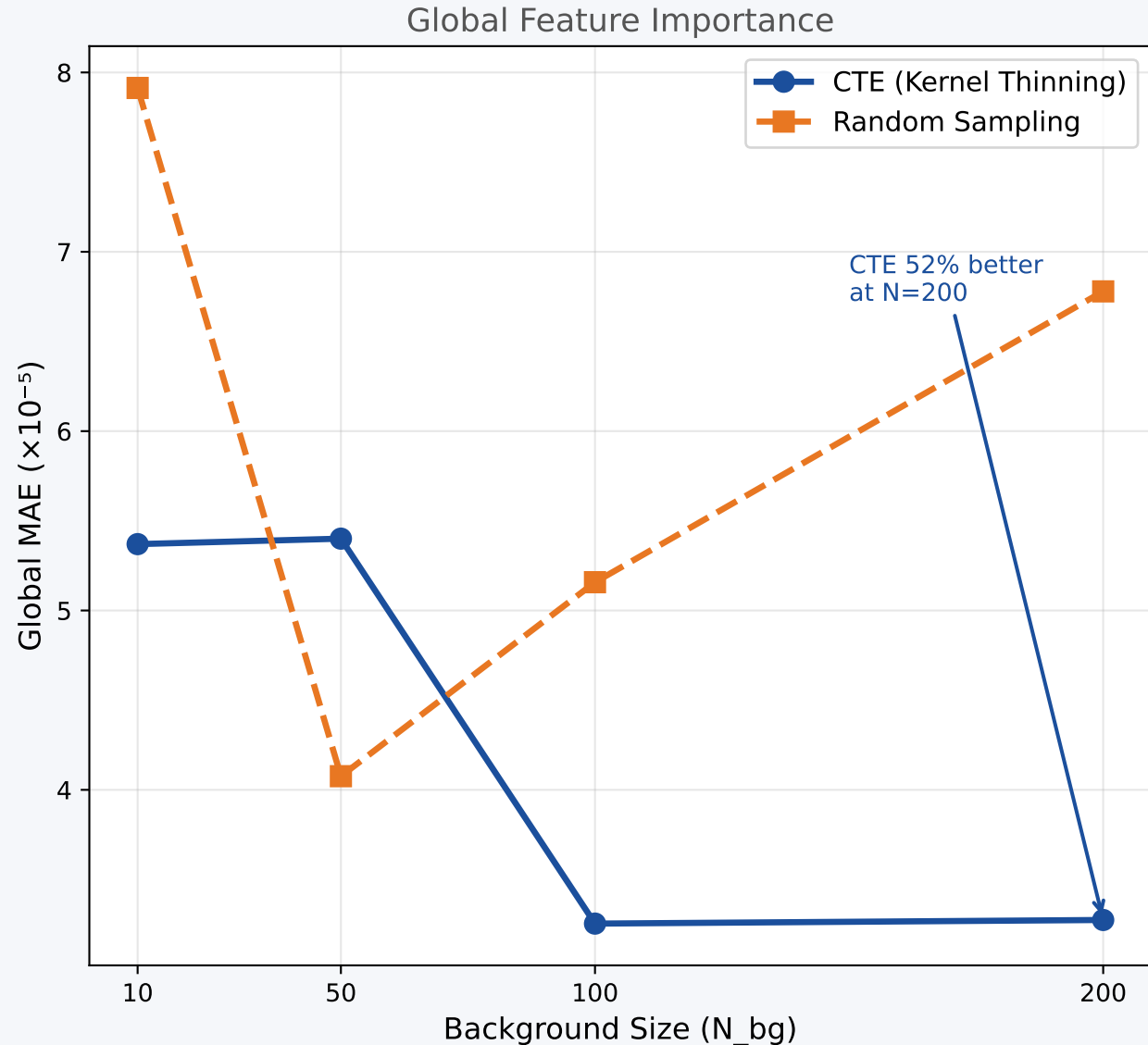
Do SHAP values look different when computed on training vs validation data? Which features drift most?

RQ4

Break-even Cost Analysis

Kernel thinning is expensive to build once. After how many batches does CTE become cheaper than re-sampling randomly?

RQ1 — Fidelity vs Background Size



Finding: CTE achieves consistently lower error than random — advantage grows with background size (up to 2× at N=200 global)

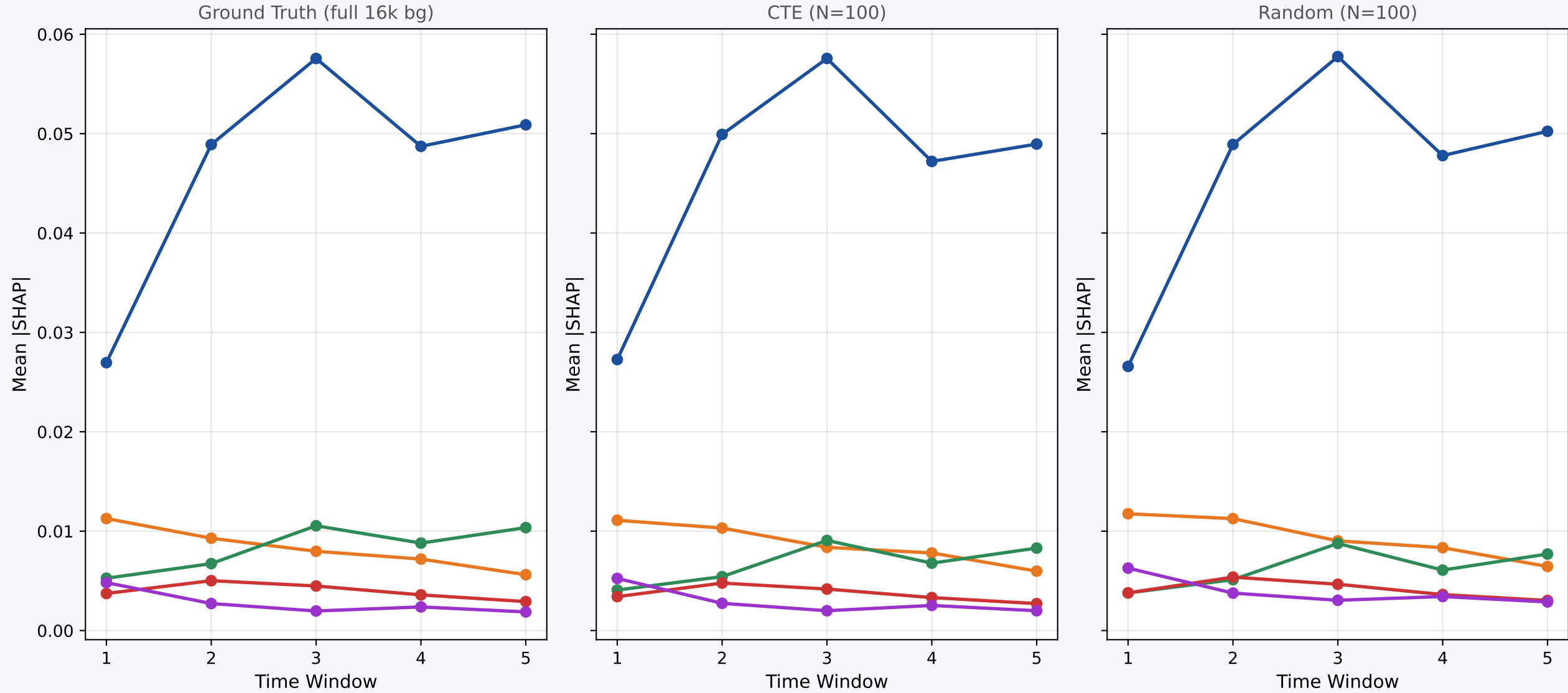
RQ1 — Detailed Results

N_bg	CTE Global MAE	Rand Global MAE	Δ Improvement	CTE Local MAE	Rand Local MAE	CTE Speedup
10	5.37×10^{-5}	7.91×10^{-5}	+32%	15.97×10^{-5}	12.68×10^{-5}	8.23×
50	5.40×10^{-5}	4.08×10^{-5}	-32%	10.37×10^{-5}	6.55×10^{-5}	1.51×
100	3.25×10^{-5}	5.16×10^{-5}	+37%	6.50×10^{-5}	7.73×10^{-5}	0.92×
200	3.27×10^{-5}	6.78×10^{-5}	+52%	6.04×10^{-5}	14.00×10^{-5}	1.00×

CTE outperforms random on Global MAE at N=10, 100, 200. Mixed result at N=50 (random slightly better).

Speedup vs ground truth (16k bg): ×8 at N=10 → ×1 at N=100 (expected: larger bg = more computation)

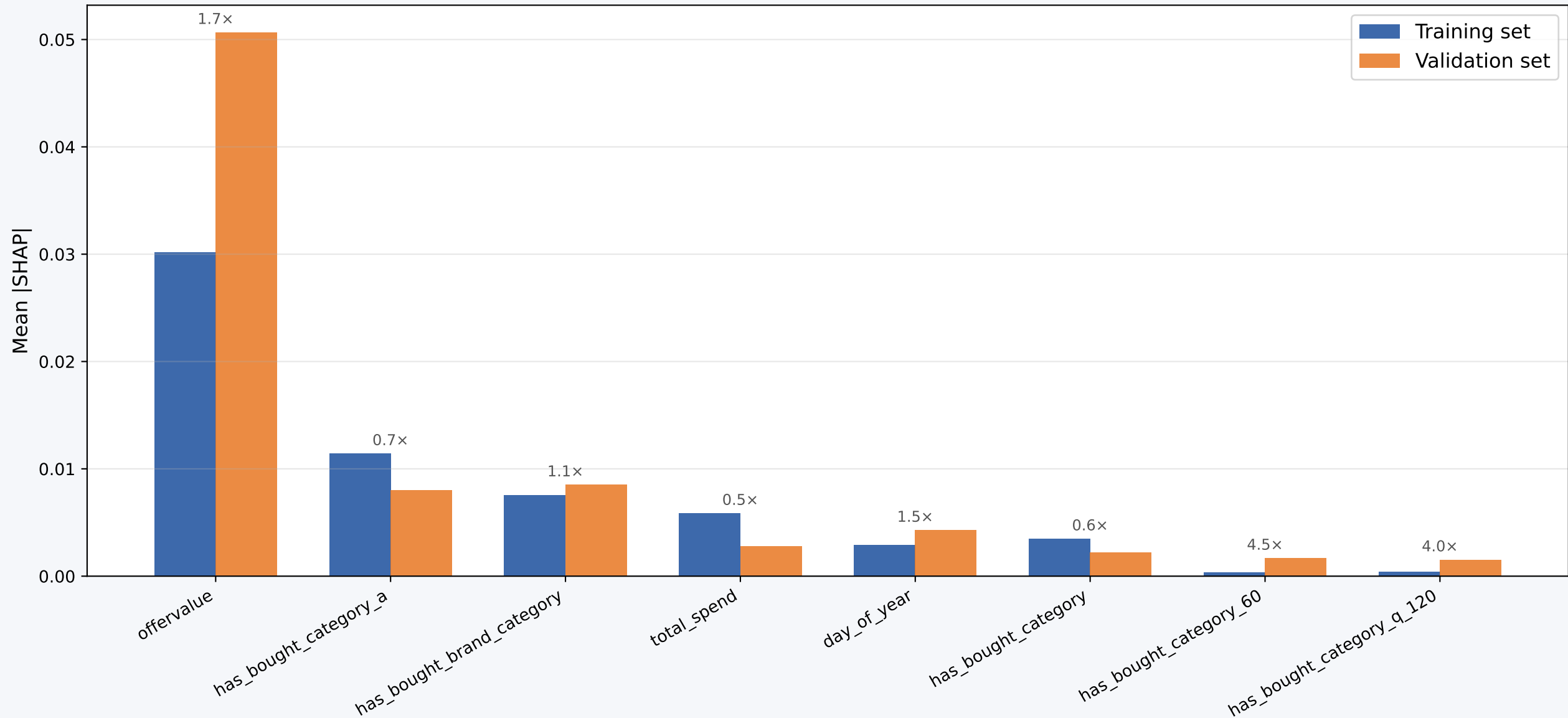
RQ2 — Feature Importance Drift Across Time Windows



Finding: offervalue importance rises then stabilises — CTE tracks the ground truth trend faithfully

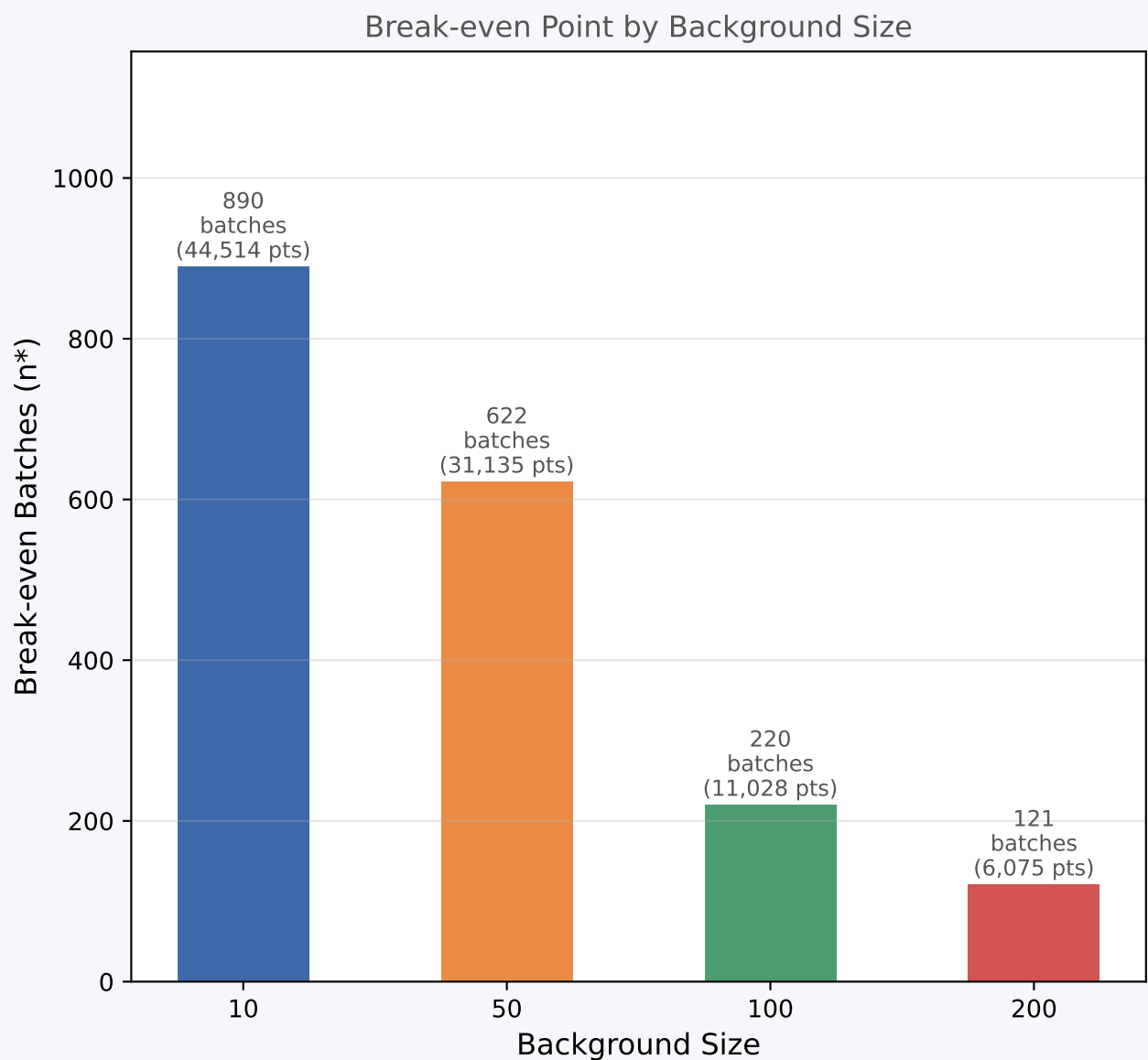
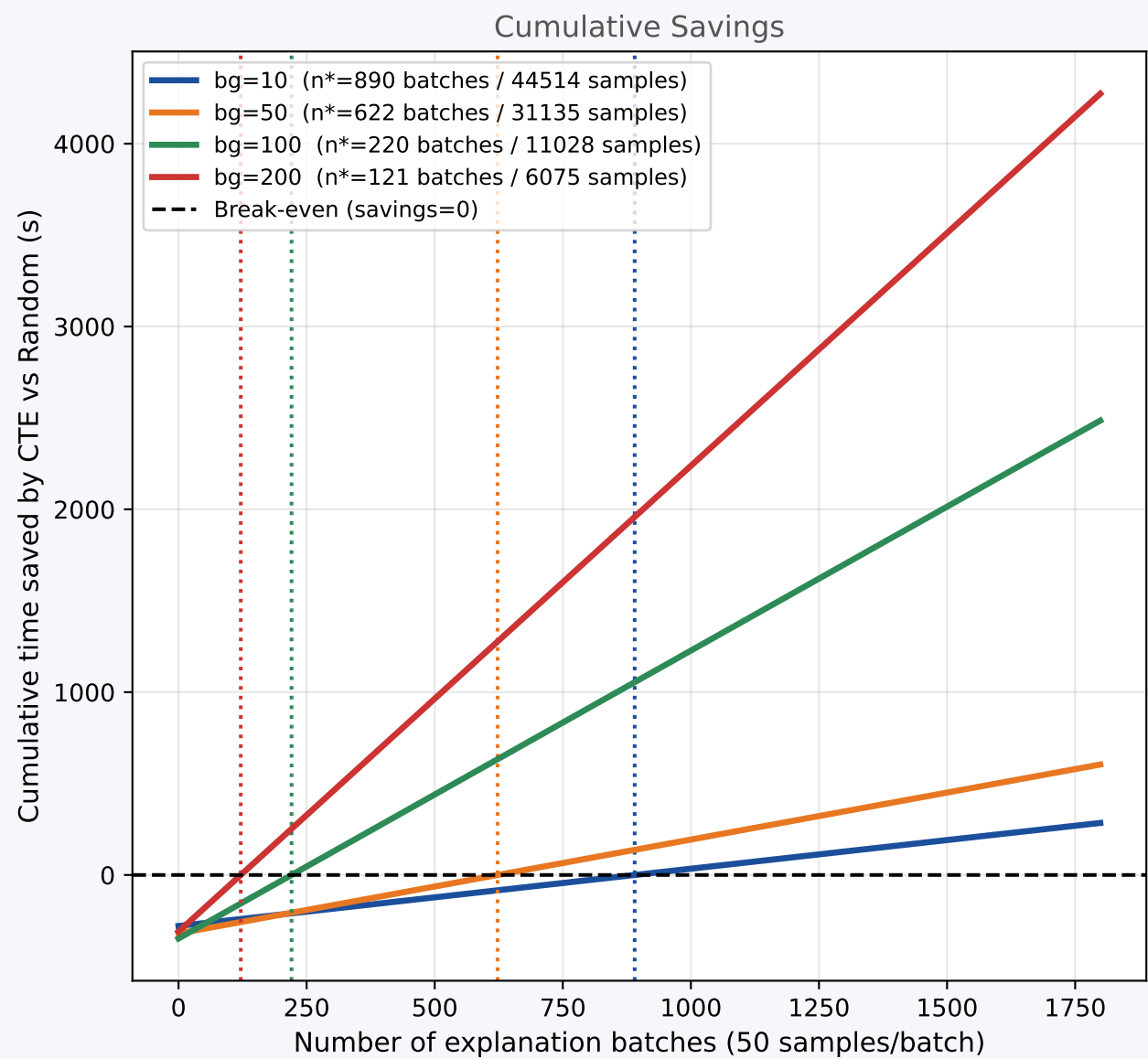
offervalue has_bought_category_a has_bought_brand_category day_of_year total_spend

RQ3 — Train vs Validation Feature Importance (Ground Truth)



Finding: offervalue is 67% more important on validation — suggests temporal distribution shift. CTE/Random closely track the ground-truth ratios.

RQ4 — CTE Break-even Analysis (KernelExplainer)



Finding: larger backgrounds break even faster. bg=200 needs only ~122 batches (~6,000 points) to recoup the CTE build cost.

Conclusions & Next Steps

✓ RQ1

CTE achieves lower global MAE than random in 3 of 4 sizes tested.
Advantage is consistent and grows with background size (up to 2× at N=200).

✓ RQ2

offervalue importance rises from window 1→3 then stabilises — temporal distribution shift detected.
CTE background (fixed, built on training data) tracks the trend as well as random.

✓ RQ3

offervalue is 67% more important on validation than train — evidence of distribution shift.
CTE and random both approximate the ground truth closely (within ~5% error).

✓ RQ4

KernelExplainer break-even: bg=200 → ~122 batches = 6,075 samples.
For production-scale usage the CTE investment pays off quickly.

Next Steps

Run on full dataset · Try LIME & permutation importance · Compare explanation methods across model families